

Strategic Partners, Academic Mentorship & Institutional Alliances

Official Executive Briefing on Ecosystem Collaborations, Scientific Advisors & Integrated Research Repositories

1. STRATEGIC & EDUCATIONAL PARTNERSHIPS

4 Active Organizations



NXT Horizon

Frontier AI Research Organization · nxthorizon.org

Featured Strategic Partner · AI Safety & Spectral Graph Theory

FEATURED STRATEGIC PARTNER

Collaborating with BioLayers AI to pioneer next-generation **spectral graph theory**, **zero-hallucination AI safety architectures**, and **multi-scale causal reasoning networks** — engineering mathematically verifiable evidence grounding that allows computational oncology graphs to extract and validate complex cancer mechanisms.

Spectral Graph Theory

Advancing directional graph algorithms, topological mechanism synthesis, and multi-scale causal networks.

AI Safety & Verification

Engineering robust, zero-hallucination mathematical architectures with verifiable evidence grounding.

Frontier AI Ecosystem

Connecting frontier AI benchmarks, open mathematical research, and precision oncology globally.

Spectral Graph Theory AI Safety AI4Math Causal Networks Zero-Hallucination



STEMise

International STEM Education Nonprofit · stemise.org

Global STEM Education Partner · Fiscally Sponsored by The Hack Foundation (501c3)

GLOBAL STEM PARTNER

International youth-led nonprofit dedicated to redefining STEM education through hands-on learning, STEM kits, and global technology programs. Collaborating with BioLayers AI to democratize complex oncology literature into visual, interactive maps for students, aspiring researchers, and STEM learners globally.

Hands-On STEM Access

Democratizing practical biomedical and computational inquiry for learners worldwide.

Open Curriculum & Workshops

Integrating interactive cancer knowledge graphs and causal diagrams into youth research programs.

Global Youth Research

Empowering the next generation of scientific leaders to explore molecular biology without barriers.

STEM Education Youth Research Open Curriculum AI Literacy Hands-On Science



CompBio Society at UCLA

UCLA Student Organization

compbiosocucla.wixsite.com/my-site

UCLA ACADEMIC PARTNER

Student-run society at the University of California, Los Angeles connecting researchers and students across Computational and Systems Biology (CaSB), bioinformatics, and genomics. Collaborating on student research seminars, technical workshops, and literature exploration.

UCLA Computational Biology Bioinformatics CaSB



BioMedizone

Global Biomedical Nonprofit

biomedizone.org · 68+ Countries

GLOBAL BIOMEDICAL PARTNER

One of the world's largest student-led biomedical nonprofits (3,000+ students, 2.5M impressions across 68 countries). Collaborating with BioLayers AI to empower youth research scholars with virtual biomedical instruction, capstones, and interactive literature tools.

Biomedical Research 68 Countries Research Scholars

2. SCIENTIFIC MENTORSHIP & AFFILIATED RESEARCH LABORATORIES

Leading Physician-Scientists & Investigators



John-William Sidhom, M.D., Ph.D.
Physician-Scientist in Medical Oncology & Computational Biology
Scientific Mentor to BioLayers AI Founder

SCIENTIFIC MENTOR

Providing core scientific mentorship to ground BioLayers AI in clinically meaningful oncology workflows, multi-omics cancer genomics, deep learning, and rigorous translational evidence.



CLINICAL FELLOWSHIP TRAINING
Weill Cornell Medicine · NewYork-Presbyterian
Hematology & Medical Oncology



DOCTORAL & MEDICAL TRAINING
Johns Hopkins University School of Medicine
M.D. · Ph.D. in Biomedical Engineering

Computational Oncology

Cancer Genomics

Deep Learning

Precision Oncology

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John Paul Shen, M.D.
Assistant Professor, Gastrointestinal Medical Oncology · MD Anderson Cancer Center
Principal Investigator, The Shen Laboratory · CPRIT Scholar

SCIENTIFIC ADVISOR

● Replied / Interested in BioLayers AI; Occasional Mentorship Requested Aug 25 — Awaiting Reply

Physician-scientist leading the **Shen Laboratory at MD Anderson**, specializing in cancer genomics, high-throughput CRISPR screening, synthetic lethal interactions, and machine learning biomarkers in colorectal and gastrointestinal malignancies.



FACULTY APPOINTMENT
MD Anderson Cancer Center
Assistant Professor, GI Medical Oncology



GRADUATE & FELLOWSHIP TRAINING
University of California, San Diego (UCSD)
Ph.D. & Medical Oncology Fellowship

The Shen Laboratory

CRISPR Screens

Synthetic Lethality

GI Oncology

linkedin.com/in/john-paul-shen-393a2ab



Eric E. Gardner, Ph.D.
Principal Investigator, James P. Allison Institute
Assistant Professor, Thoracic/Head & Neck Medical Oncology · MD Anderson

SCIENTIFIC GUIDANCE

● Occasional High-Level Guidance Offered; Project Direction Requested Aug 25 — Awaiting Reply

Principal Investigator at the prestigious **James P. Allison Institute** and MD Anderson Department of Thoracic/Head & Neck Medical Oncology, investigating cancer cell plasticity, airway damage, tumor immune evasion, and novel therapeutic targets in thoracic malignancies.



FACULTY & INSTITUTE APPOINTMENT
James P. Allison Institute · MD Anderson
Assistant Professor, Thoracic Oncology



DOCTORAL TRAINING
Johns Hopkins University School of Medicine
Ph.D. Graduate Training

James P. Allison Institute

Thoracic Oncology

Cancer Plasticity

linkedin.com/in/eric-edison-gardner-7b44a8120

* Institutional names and affiliations (MD Anderson Cancer Center, James P. Allison Institute, Weill Cornell Medicine, NewYork-Presbyterian Hospital, Johns Hopkins University School of Medicine, UC San Diego) are presented solely as biographical and academic context for individual scientists and mentors, and do not imply institutional sponsorship or endorsement of BioLayers AI.

3. GLOBAL RECOGNITION & SPOTLIGHTS

International Media & EdTech Catalysts

HundrED Global Innovation

HELSINKI · 17+ COUNTRIES

Selected and featured by **HundrED (Helsinki, Finland)** in their Global Education Innovation catalog for democratizing cancer research into visual, explorable knowledge graphs across 17+ countries.

hundred.org/en/innovations/biolayers-ai

AI Business Feature

STARTUP & FOUNDER SPOTLIGHT

Featured in **AI Business**: "A teenager in Tashkent read 100 cancer papers and built the tool he was missing — transforming fragmented literature into explorable mechanistic maps."

aibusiness.vc/startups/biolayers-ai

4. LIVE BIOMEDICAL DATA & ONTOLOGY INTEGRATIONS

Production API Endpoints

NCBI / PubMed (NIH)

Live integration via NCBI E-utilities (/api/pubmed) for biomedical literature extraction, abstract parsing, citation mapping, and PMCID full-text resolution.

EMBL-EBI / Cell Ontology (OLS)

Live integration with the European Bioinformatics Institute Ontology Lookup Service (/api/cells) for standardized cell-type taxonomy and hierarchical validation.